Typst Workshop

The Modern Typesetting Solution

Rishi Vora

Part 1: Why Typst?

What's wrong with LaTeX??

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- **The Gold Standard**: De facto standard for academic and technical publishing since the 80s.

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- **The Gold Standard**: De facto standard for academic and technical publishing since the 80s.
- **Legacy Baggage**:
  - **Heavy Installation**: 2GB+ distributions (TeX Live, MiKTeX).
  - **Slow Feedback**: Save → Wait 5-10s → Check PDF.
  - **Opaque Errors**: "Undefined control sequence" in a package you didn't know you imported.
  - **Programming in TeX**: Macros are powerful but notoriously difficult to debug and maintain. Not to mention the plethora of backslashes and curly braces!

Typst: An Update to Typesetting

Typst is a new, open-source typesetting system that aims to be as powerful as LaTeX while being much easier to learn and use.

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Feature	LaTeX	Typst
Speed	Slow (takes seconds)	Instant (milliseconds)
Language	Macro-based (TeX)	Native Scripting
Installation	Heavy (2GB+)	Single Binary (40MB)
Packages	Global/Fragmented (CTAN)	Managed (Typst Universe)
Error Handling	Cryptic / Stack-heavy	Clear / Visual

The Ecosystem: Typst Universe

Typst uses a modern package management system similar to `uv` or `cargo`.

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Typst uses a modern package management system similar to `uv` or `cargo`.

- **No more manual downloads**: Just import with `#import "@preview/package:version"`.
- **Versioned Dependencies**: Ensures your document compiles the same way 5 years from now.
- **Central Registry**: Explore themes, templates, and packages at typst.app/universe.

Adoption

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Typst is powerful, but it's still a very new language!

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Current Reality:

- **Journals**: Most publishers still require LaTeX source.
- **Conferences**: Adoption is starting, but LaTeX remains the default.

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The Good News:

- **arXiv**: Native support for Typst is officially coming soon!

Where Typst Shines TODAY

You don’t have to wait for every journal to switch to start using Typst.

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You don't have to wait for every journal to switch to start using Typst.

Presentations:

Using libraries like `touying`, `diatypst`, or `grape-suite` (like this one!).

CVs & Resumes

Rishi Vora

Final Year BS-MS Student in Physics at IISER Mohali

vorarishi22@gmail.com | ms21113@iisermohali.ac.in | Portfolio | GitHub | LinkedIn

Education

2021 – 2026 | **Indian Institute of Science Education and Research (IISER),** Mohali
Integrated Bachelors & Masters in Physics, with Minor in Data Science

Relevant Courses: Foundations of Quantum Mechanics and Quantum Information | Advanced Quantum Mechanics | Machine Learning and Artificial Intelligence | Introduction to Data Science | Algorithms and Complexity | Computational Methods in Physics | Modelling Complex Systems | Nonlinear Dynamics, Chaos, and Complex Systems | Theory of Computation

CPI: 7.6 / 10 (upto 9th Semester)

2021 | **eduNova Science Higher Secondary School,** Ahmedabad
12th - Gujarat Secondary and Higher Secondary Education Board

Grade: A2
Percentile: 92.25

Research Experience

06/2025 – Present | **Generative quantum machine learning**
IISER Pune

Guide: Prof. Santhanam

- Developed a modular, high-performance **Generative QML** framework in **Julia** (Yao.jl) using trainable **Parametrized Quantum Circuits** (PQCs) to synthesize quantum ensembles.
- Optimized the **Quantum DDPM** prototype by Zhang et al., achieving a **10-15x training speedup**.
- **Benchmarked** hybrid optimization strategies, including from-scratch implementations of gradient-free methods (**QNSPSA**, **SPSA**, **Rotosolve**) against gradient-based approaches (Adam, AMSGrad).
- Exploring **Quantum Optimal Transport** (QOT) metrics such as **Quantum Wasserstein distance** as an alternative for training generative models.
- Scaling generative capabilities to **10-qubit systems** to study quantum chaos and localization in the **Quantum Kicked Rotor**.

Summer 2024 | **Reading project on QKD protocols**
IISER Mohali

Guide: Prof. Arvind

- Reviewed working of BB84, B92 and E91 QKD protocols.
- Worked out Shor and Preskill's security proof of BB84 protocol.

Summer 2023 | **Quantum algorithms and circuit decomposition**
IISER Mohali

Guide: Dr. Sandeep Goyal

- Used Qiskit to implement basic circuits like Full Adder, Multiplier, Superposition Adder, and XOR adder.
- Studied Deutsch-Jozsa and Shor's algorithm.
- Attempted a circuit decomposition scheme using Solovay-Kitaev algorithm.

Skills & Certifications

2024 | **Qiskit Global Summer School 2024**
IBM

- Certificate of Quantum Excellence
- Lectures and Interactive Sessions on Circuit Transpilation, Modeling and Characterization of Hardware Noise, Execution on Noisy Quantum Hardware using Qiskit.

2023 | **Qiskit Global Summer School 2023**
IBM

- Certificate of Quantum Excellence
- Lectures and Interactive Sessions on Dynamic circuits, Phase estimation, Factoring, Error mitigation, Variational Quantum Eigensolver and Iterative phase estimation.

Programming & Systems | **Python** (Qiskit, PyTorch, TensorCircuit) | **Julia** (Yao.jl) | **Linux** (NixOS)

Tools & Frameworks | **Git** / **GitHub** | **Typst**

Languages | Fluent in **English**, **Hindi**, as well as **Gujarati**

Hobbies | Maintaining and contributing to FOSS projects on **GitHub** | Web Development | Listening to music | Watching eclectic films

Scholarships

2021 – Present | **IISER Mohali**
Merit-Cum-Means Scholarship

Bi-annual scholarship supporting Bachelor's and Master's studies.

Extracurriculars

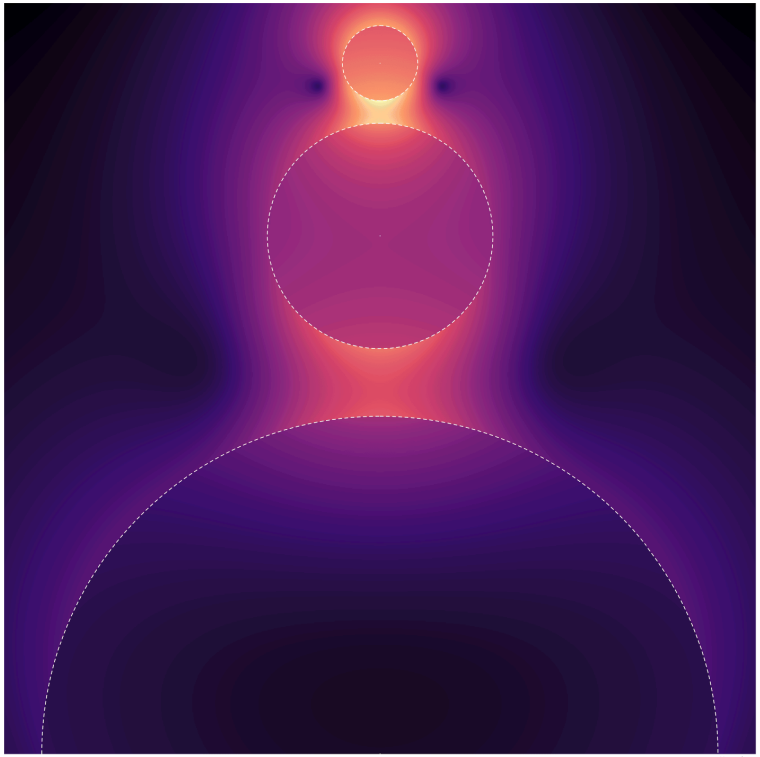
March 2026 | **Conducted Workshops on Python Tooling, Julia, and Typst**
IISER Pune

- **Link**: https://github.com/Vortriz/workshops
- Conducted a series of interactive workshops focused on addressing problems with **Python tooling** ecosystem, introducing **Julia** as a performant alternative for scientific computing, and showcasing **Typst** for high-quality document preparation.

2023 – 2024 | **Convenor of Turing Club**
IISER Mohali

Organized Hackathons, Workshops, Talks etc. Took programming help sessions for junior batches.

POSTERS



Nano-lens

T · E · R · M · S

MULTIPLE SCATTERING OF LIGHT

Dmitri Scherbarchov
Atefeh Fazel-Najafabadi
Eric Le Ru
Baptiste Auguié

nano-optics.ac.nz/terms
2025-07-19

TERMS stands for T-matrix for Electromagnetic Radiation with Multiple Scatterers — it is a Fortran program to simulate the near-field and far-field optical properties of collections of particles. TERMS solves rigorously the Maxwell equations via the superposition T-matrix method, where incident and scattered fields are decomposed into a basis of multipolar electric and magnetic spherical waves.

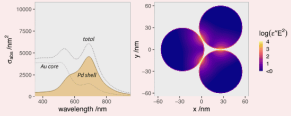
In a multiple-scattering problem the net field exciting a given particle is composed of the incident field plus the scattering contribution from neighbouring particles, resulting in a coupled system of equations to be solved for the total fields.

TERMS implements several algorithms to describe the self-consistent electromagnetic interaction between multiple scatterers, and from there compute optical properties such as absorption, scattering, extinction, circular dichroism, as well as near-field intensities and the local degree of optical chirality.

By describing the incident and scattered fields in a basis of spherical waves the T-matrix framework lends itself to analytical formulas for orientation-averaged quantities such as far-field cross-sections and near-field intensities, greatly reducing the computational time needed to simulate particles and systems of particles in random orientation.

EXAMPLE INPUT FILE

```
> ModeAndScheme 2 3
> MultipoleCutoff 15
> Wavelength 300 1000 300
> Medium 1.7689 # water
> Incidence 0.0 0.0 0.0 1
> # trimer of core-shell spheres
> Scatterers 3
> Au@Pd  0.0      0.0 0.0 30.0 29.0
> Au@Pd 52.481 30.3 0.0 30.0 29.0
> Au@Pd 52.481 30.3 0.0 30.0 29.0
```



Results from a simulation using the above input script

TERMS AND CONDITIONS

TERMS requires a Fortran 90 compiler; it processes input text files with specific keywords controlling the simulation parameters. Options include,

1. Far-field quantities (absorption, scattering, extinction, circular dichroism) for multiple wavelengths and angles of incidence, as well as orientation-averages
2. Near-field calculations for multiple wavelengths and incident angles, also computing the local degree of chirality, as well as orientation-averages
3. Stokes parameters and differential scattering cross-sections for multiple incidence or scattering angles

The computational cost scales with the size of the linear system, proportional to the number of particles N_p, and to the square of the maximum multipolar order N_{max}. On a typical PC we may treat up to ~ 500 particles with $N_{max} = 1$, and a dimer with N_{max} up to ~ 60.

THE T-MATRIX METHOD

Incident and scattered fields are expanded in a basis of Vector Spherical Waves (VSWs) $\boldsymbol{M}, \boldsymbol{M}$,

$$\boldsymbol{E}_{inc}(\boldsymbol{r}) = \sum_{p=1}^{2} \sum_{n=1}^{n_{max}} \sum_{m=-n}^{n} c^p_{nm} \boldsymbol{M}^p_{nm}(\boldsymbol{r})$$

$$\boldsymbol{E}_{sca}(\boldsymbol{r}) = \sum_{p=1}^{2} \sum_{n=1}^{n_{max}} \sum_{m=-n}^{n} c^p_{nm} \boldsymbol{M}^p_{nm}(\boldsymbol{r})$$

where n is the multipolar order, $|m| \leq n$, and p is the parity (electric or magnetic type). Linearity of the Maxwell equations implies a linear relationship

$$\boldsymbol{c} = \boldsymbol{T}\boldsymbol{c}$$

defining the T-matrix for one scatterer. In practice, all quantities are truncated to a maximum multipolar order N_{max}.

MULTIPLE SCATTERING

Where multiple scatterers are present, the scattered field of one contributes to excite the others, leading to a coupled system of equations,

$$\boldsymbol{c}^{(i)}_{exc} = \boldsymbol{c}^{(i)}_{inc} + \sum_{\substack{j=1 \\ i \neq j}}^{N} \boldsymbol{O}^{(i,j)} \boldsymbol{T}^{(i)}_1 \boldsymbol{c}^{(i)}_{exc}$$

to be solved self-consistently for the unknown excitation coefficients $\boldsymbol{c}^{(i)}_{exc}$, with numerical linear algebra routines.

The matrices $\boldsymbol{O}^{(i,j)}$ transform the VSWs expansion centred on one scatterer to another origin, and require special care to retain accuracy.
The T-matrix method finds limitations when the VSW series fails to converge in some regions near nonspherical particles, or when the circumscribed spheres of adjacent particles intersect.

OBTAINING T-MATRICES

The T-matrix for a given particle shape and composition can be obtained by a number of techniques, including general numerical solvers of Maxwell's equations. The extended boundary-condition method is arguably the most efficient and accurate; it connects the incident, scattered and internal fields at the particle boundary and provides the tmatrix elements as surface integrals of the VSWs. Simplifications occur for axi-symmetric particles such as spheroids, and for spheres in particular the tmatrix is simply diagonal, with analytical expressions for all elements in terms of spherical Bessel functions.

CALCULATION RESULTS

With knowledge of the scattered field coefficients one can compute the fields $\boldsymbol{E}, \boldsymbol{B}$ everywhere in space, as well as $|\boldsymbol{E}|^2, |\boldsymbol{E}|^4, \mathscr{C} \propto \Im(\boldsymbol{E}^* \cdot \boldsymbol{B}), \ldots$
Far-field cross-sections for extinction, scattering, absorption are obtained as

$$\sigma_{ext} = -\frac{1}{k^2_{med}} \Re\left\{ \sum_{j=1}^{N} \boldsymbol{a}^{(j)\dagger} \boldsymbol{f}^{(j)} \right\}$$

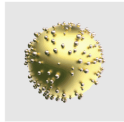
$$\sigma_{sca} = \frac{1}{k^2_{med}} \Re\left\{ \sum_{j=1}^{N} \sum_{i=1}^{N} \boldsymbol{f}^{(j)\dagger} tilde\boldsymbol{O}^{(j,i)} \boldsymbol{f}^{(i)} \right\}$$

$$\sigma_{abs} = \sigma_{ext} - \sigma_{sca}.$$

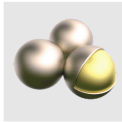
Orientation-averaged values such that

$$\langle X \rangle = \frac{1}{4\pi} \int_0^{\pi} \int_0^{2\pi} X(\varphi, \theta) \sin\theta d\varphi d\theta$$

for cross-sections $\langle \sigma_{ext} \rangle, \langle \sigma_{sca} \rangle, \langle \sigma_{abs} \rangle$ as well as near-field quantities $\langle |\boldsymbol{E}|^2 \rangle, \langle \mathscr{C} \rangle$ can be obtained analytically in terms of the T-matrices and translation matrices.

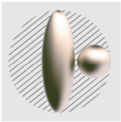


CORE-SATELLITES



AU@PD TRIMER

AU NANOROD HELIX



RAYLEIGH HYPOTHESIS

AU NANOSTAR

NOTABLE FEATURES OF THE CODE

- Incident plane waves along arbitrary directions, with linear or circular polarisation
- Built-in calculation of individual for coated spheres; import of general from other programs
- Built-in dielectric functions for common materials such as Au, Ag, Al, Cr, Pt, Pd, Si, and Water, or from tabulated values
- Per-layer absorption in layered spheres
- Orientation-averaging of far-field cross-sections, as well as linear and circular dichroism
- Near-field maps of electric and magnetic field components, $|\boldsymbol{E}|^2, |\boldsymbol{E}|^4, \mathscr{C} \propto \Im(\boldsymbol{E}^* \cdot \boldsymbol{B})$
- Calculation of the global cluster
- "Masking" of specific multipolar orders
- Calculation of Stokes parameters, phase matrix, differential scattering
- Plain text or HDF5 I/O format
- Possible compilation in quad-precision

FUTURE DEVELOPMENTS

Additions only requiring additional code

- Built-in spheroidal particles
- Internal fields of spheroidal particles
- Orientation-averaged internal fields
- Dipolar emitter

Code optimisations or improvements

- Iterative solver, preconditioner
- Geometry optimisation (via external libraries)

Improvements to the theoretical framework

- Plane-wave expansion for particles with intersecting circumscribed spheres
- Conversion from / to cartesian polarisability tensors

We may also port the codebase to the Julia language for future development. Contact us if interested! (baptiste.auguie@vuw.ac.nz)

REFERENCES

T-matrix method and algorithms used in the code

1. M. Mishchenko, L. Travis, and A. Lacis, Camb. U. Press (2002)
2. B. Stout, J.-C. Auger, and J. Lafait, J Mod Opt, 49 (2002)
3. B. Stout, J. C. Auger, and A. Devilez, JOSA A, 25 (2008)
4. D. Mackowski Proc. Royal Soc. A, 433 (1991)
5. D. Mackowski, JOSA A, 11 (1994)
6. D. Mackowski and M. Mishchenko, JOSA A, 13 (1996)
7. D. Mackowski and M. Mishchenko, JQSRT, 112 (2011)
8. R. N. S. Suryadharma and C. Rockstuhl, Materials (2018)

Publications using TERMS:

1. D. Schebarchov, E. C. Le Ru et al. Opt. Express 27 (2019)
2. S. Lee, H Hwang et al, ACS Energy Lett, 3881–3890 (2020)
3. A. Fazel-Najafabadi, S. Schuster et al. Phys. Rev. B 103 (2020)
4. A. Fazel-Najafabadi, B. Auguié Mat Adv 3 (2021)
5. M. Herran, A. Sousa-Castillo et al. Adv Func Mat 32, (2022)
6. A. Fazel-Najafabadi, B. Auguié JQSRT 286 (2022)
7. D. Schebarchov, A. Fazel-Najafabadi et al. JQSRT 284 (2022)
8. L. Velasco, T. Ouyang, B.M. Reinhard Nano Lett. 23 (2023)
9. S. Glukhova, E.C. Le Ru, B. Auguié Nanoscale 15 (2023)
10. T. Ouyang, Y-C. Chen et al. ACS Nano 18 (2024)
11. [...] E.C. Le Ru, B. Auguié et al. JQSRT 333 (2025)

Reports & Theses

Chapter 2

Exoplanets, a primer

This thesis sits right at the interface of two different disciplines: Machine learning (i.e., computer science) on the one hand, and exoplanet science (i.e., astrophysics) on the other. To appreciate its contents, a certain understanding of both sides is useful. The main parts in chapters 3, 4, and 5 are written in a way that assumes that the reader is familiar with the astrophysical context, and focuses more on explaining the methodical innovations that are being introduced. To make things more accessible to people without a strong background in astronomy, this chapter offers a first introduction to the field of exoplanet science. Its goal is to provide an overview of selected concepts and ideas that set the stage on which the main contributions of this thesis take place. It can also be seen as an (incomplete) summary of things I have learned during the course of my Ph.D. studies and have found useful in my interactions with exoplanet scientists. Of course, given the vastness of the subject, this introduction can only scratch the surface and will be by no means complete or comprehensive. For this reason, an extensive bibliography is provided, which can serve as a starting point for the interested reader. Moreover, I advert to the books by Seager (2010b), Perryman (2018), and Deeg & Belmonte (2018), who provide in-depth reviews of virtually every aspect of exoplanet science.

Contributions
I have written and edited this chapter based on my understanding of the subject and my review of the literature. Unless explicitly noted otherwise, all figures have been created by me.

Advice for reading this chapter. *This chapter is written such that the individual sections at least partially build on each other. Nevertheless, if a reader finds themselves pressed for time, I recommend focussing on sections 2.4 (High-contrast imaging) and 2.5.2 (Atmospheric retrieval), which are most directly relevant to the main part of this thesis.*

2.1 WHAT IS AN EXOPLANET, ANYWAY?

"I work on machine learning for astrophysics, specifically exoplanets. Um, do you know what an exoplanet is?" —— This exact sequence of words is how many conversations have started for me over the past few years. More often than not, the answer was but a slightly abashed smile, which has led me to believe that perhaps it would only be appropriate to

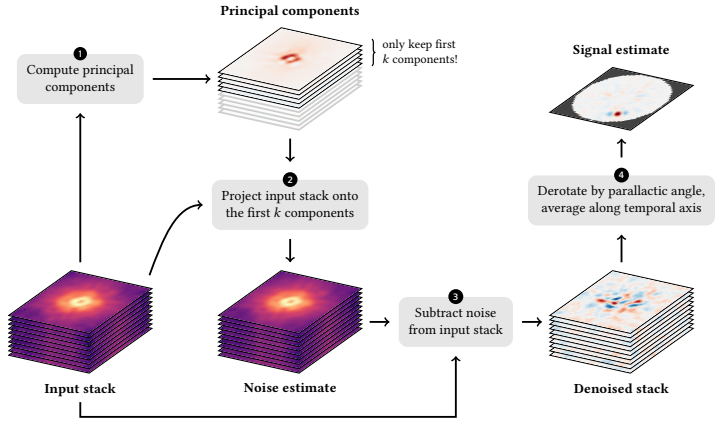


Figure 2.19: Schematic illustration of PCA-based PSF subtraction for ADI. For presentation purposes, different parts of the figure use different color maps. The differences between the input stack and the noise estimate are not visible to the naked eye.

is still widely used both for new discoveries (see, e.g., Mesa et al., 2023, for a recent example) and as a baseline for method development. Second, the algorithm is remarkably simple and easy to explain. Its basic principle is illustrated schematically in fig. 2.19. We outline the application to ADI data here, but transferring the main idea to RDI is straightforward.

The algorithm starts with the "input stack" (i.e., the pre-processed sequence of images) and computes a basis of principal components (or "eigenimages") for it.[63] The principal components (PCs) can be computed efficiently using a singular value decomposition (SVD) assuming that the data are centered, that is, the temporal mean over all frames has been subtracted from each frame.[64] Next, the basis of PCs is truncated to the first k components (in descending order of explained variance). The value of k is the main hyperparameter of the PCA/KLIP algorithm. Its choice depends on the dataset, as well as on the separation of a companion from its host. In the second step, the original input stack is projected onto the PCs, that is, each frame is expressed in the truncated PC basis. This corresponds to the optimal linear dimensionality reduction to k dimensions.[65] The output of this step is our estimate of the PSF. Note that, unlike in median ADI, we get a different estimate for each frame in the input stack. In the next step, the PSF estimate is subtracted from the input stack, and the result is derotated and averaged. Using a

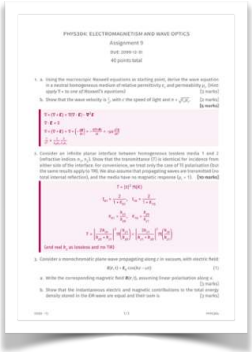
[63] When relying on existing PCA implementations, this may require the images to be flattened into 1D vectors.

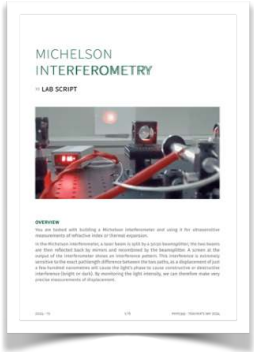
[64] Note that this step can already introduce a bias if the mean contains part of the planet signal.

[65] "Optimal" in the sense that this projection minimizes the L_2-distance between the original data and their low-rank approximation; see, e.g., Cunningham & Ghahramani (2015).

And much more...

EXAM


ASSESSMENT



LAB SCRIPT

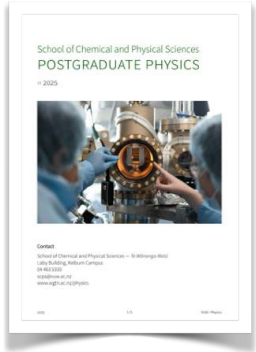
PRESENTATION

MARKING SHEET

COURSE INFO

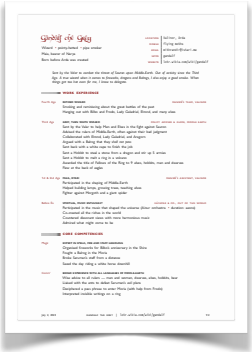
COURSE POSTER

PROGRAMME



BOOKLET

BANNER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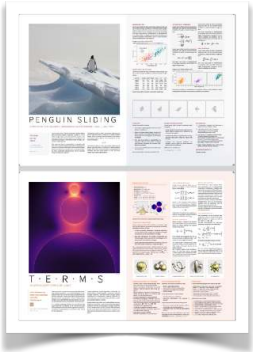
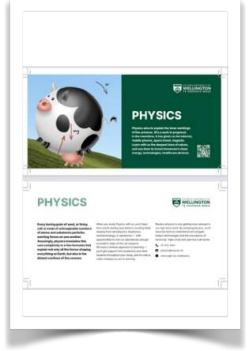


RESEARCH POSTER

FLYER

BROCHURE



POSTCARD

LEAFLET

AGENDA

CALENDAR

Part 2: Demo

Quick Links

Downloading Typst for Local Use

- https://typst.app/open-source/#download
- Or use Tinymist VS Code extension.

Documentation & Learning Resources

- https://typst.app/docs/
- https://typst.app/docs/guides/for-latex-users/
- https://typst.app/docs/reference/syntax/
- https://typst.app/docs/reference/symbols/sym/

Packages & Templates

- https://typst.app/universe/

Thank you!

Questions?